

Scenario 1 output 1 ollama

Here is the detailed threat modelling analysis for the LLM-Integrated Authentication Assistant System:

****STRIDE-based Threats****

1. ****Threat ID:**** STR01

****Threat Name:**** Unauthorized Access

****STRIDE Category:**** T (Tampering)

****Affected Component/Data Flow:**** P2, User Database, Session Storage

****Attack Scenario:**** An attacker submits a malicious login request to bypass authentication.

****Security Impact:**** Compromise of user credentials and potential access to sensitive data.

****Severity:**** High

****Likelihood:**** Medium

****Mitigation Strategy:**** Implement robust authentication mechanisms in P2

and ensure secure communication with the User Database and Session Storage.

2. ****Threat ID:**** STR02

****Threat Name:**** Data Tampering

****STRIDE Category:**** T (Tampering)

****Affected Component/Data Flow:**** P3, Session Management Module, Session Storage

****Attack Scenario:**** An attacker manipulates session data to gain unauthorized access.

****Security Impact:**** Compromise of session security and potential access to sensitive data.

****Severity:**** High.,/.

****Likelihood:**** Medium

****Mitigation Strategy:**** Implement secure session management in P3 and ensure proper validation of session data in Session Storage.

****LLM-specific Security Threats****

1. ****Threat ID:**** LLM01

****Threat Name:**** Malicious LLM Output

****Affected Component/Data Flow:**** LLM Integration Service, External LLM Provider

****Attack Scenario:**** The external LLM provider generates malicious output

that is not caught by the application-level response handling.

****Security Impact:**** Compromise of system integrity and potential unauthorized access.

****Severity:**** High

****Likelihood:**** Low

****Mitigation Strategy:**** Implement proper validation and sanitization of LLM output in the LLM Integration Service.

2. ****Threat ID:**** LLM02

****Threat Name:**** Prompt Injection

****Affected Component/Data Flow:**** P5, LLM Integration Service, External LLM Provider
****Attack Scenario:**** An attacker injects malicious prompts to manipulate the external LLM's behavior.
****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Medium
****Mitigation Strategy:**** Implement robust input validation in P5 and ensure proper sanitization of prompts before sending them to the External LLM Provider.

****OWASP LLM Top 10 Relevant Threats****

1. ****Threat ID:**** OWASP01

****Threat Name:**** Insecure Input Validation
****Affected Component/Data Flow:**** P1, Web Frontend Interface
****Attack Scenario:**** An attacker submits malicious input that is not caught by the application-level validation.
****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Medium
****Mitigation Strategy:**** Implement robust input validation in P1 to prevent malicious input.

2. ****Threat ID:**** OWASP02

****Threat Name:**** Sensitive Information Disclosure
****Affected Component/Data Flow:**** P2, User Database
****Attack Scenario:**** An attacker gains unauthorized access to sensitive user information.
****Security Impact:**** Compromise of user data and potential identity theft.
****Severity:**** Critical
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust access controls in P2 and ensure proper encryption of sensitive user data.

3. ****Threat ID:**** OWASP03

****Threat Name:**** Insecure Output Handling
****Affected Component/Data Flow:**** LLM Integration Service, External LLM Provider
****Attack Scenario:**** The external LLM provider generates output that is not properly validated or sanitized by the application.
****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Low
****Mitigation Strategy:**** Implement proper validation and sanitization of LLM output in the LLM Integration Service.

****Threats related to Trust Boundaries****

1. ****Threat ID:**** TB01

****Threat Name:** Trust Boundary Crossing (Data)**
****Affected Component/Data Flow:** External User Environment, Application Environment**
****Attack Scenario:** An attacker submits malicious input that crosses the trust boundary and compromises system integrity.**
****Security Impact:** Compromise of system integrity and potential unauthorized access.**
****Severity:** High**
****Likelihood:** Medium**
****Mitigation Strategy:** Implement robust input validation in P1 to prevent malicious input from crossing the trust boundary.**

2. ****Threat ID:** TB02**

****Threat Name:** Trust Boundary Crossing (Session)**
****Affected Component/Data Flow:** Session Storage, External LLM Provider**
****Attack Scenario:** An attacker manipulates session data that crosses the trust boundary and compromises system integrity.**
****Security Impact:** Compromise of session security and potential unauthorized access.**
****Severity:** High**
****Likelihood:** Medium**
****Mitigation Strategy:** Implement secure session management in P3 and ensure proper validation of session data before sending it to the External LLM Provider.**

****Threats related to Prompt Injection, Sensitive Information Disclosure, Insecure Output Handling, Excessive Agency, Session Security, Authentication Abuse, Data Leakage****

These threats are already covered by the above-mentioned STRIDE-based, LLM-specific, and OWASP LLM Top 10 relevant threats.